Supplementary Material

Supplementary Table S1: Group-level metric summaries across major metric families. Summary values are shown for the Pre-LLM, LLM-independent, and LLM-assisted groups. Values are means with standard deviations. Available sample sizes vary by metric.

Metric	Pre-LLM ($n = 53,575$)	LLM-independent ($n = 12,961$)	LLM-assisted ($n = 2,673$)
Focal topics			
Per-paper topic counts	5.69 (1.80)	5.36 (1.64)	5.24 (1.59)
Mean distance from topic centroid	8.68 (0.95)	8.39 (0.96)	8.12 (1.05)
Reference			
Number of references	32.80 (22.44)	34.32 (22.34)	36.15 (25.97)
Reference-topic Rao-Stirling diversity	0.356 (0.189)	0.351 (0.188)	0.393 (0.179)
Focal-reference Rao-Stirling distance	0.628 (0.120)	0.610 (0.117)	0.620 (0.109)
Grassroots globalism			
Non-Anglophone, %	52.5 (49.9)	59.8 (49.0)	72.7 (44.6)
Low- or middle-income, %	3.2 (17.6)	5.4 (22.7)	5.6 (22.9)
International collaboration, %	17.1 (37.7)	18.2 (38.6)	19.9 (39.9)
Visibility			
Cited-by count	53.32 (360.46)	9.03 (85.58)	4.20 (22.69)
Cumulative citations within two years	10.08 (51.00)	12.46 (74.41)	15.53 (50.67)
Journal citation impact	3.05 (10.20)	3.12 (2.82)	3.82 (3.09)

Supplementary Table S2: Cumulative citations received at fixed observation windows. Only papers with observable follow-up through the stated month were included. Tests compare LLM-assisted with LLM-independent papers. Adjusted q values account for the three cohort comparisons within each window.

Months	Cohort	Papers	Mean	Median	p	q
12	LLM-independent	9,949	3.88	1	0.001	0.001
12	LLM-assisted	1,530	3.59	2		
24	LLM-independent	4,508	12.46	4	0.817	0.835
24	LLM-assisted	272	15.53	4		
36	LLM-independent	1,453	27.89	7	0.373	0.583
36	LLM-assisted	26	41.08	8		

Supplementary Table S3: Journal access models across publication cohorts. Access status was identifiable for 62,847 of 69,209 papers, representing 90.81% of the cohort. Gold Open Access includes fully open-access journals, including fee-free diamond journals. Percentages use papers with identifiable status within each cohort as the denominator.

Cohort	Identifiable status	Gold OA %	Subscription/hybrid %
Pre-LLM	48,515	34.87	65.13
LLM-independent	11,959	39.07	60.93
LLM-assisted	2,373	42.90	57.10

Prompt 1: Full rewrite

```
System:
You are a scientific writing assistant. Rewrite scientific abstracts while preserving
the underlying claims, methods, results, and limitations.

User:
Rewrite the following scientific abstract fully in new wording.
Preserve the scientific meaning, structure, and approximate length.
Do not add new facts, remove findings, or change the study conclusions.

Original abstract:
{abstract}

Rewritten abstract:
```

Prompt 2: Polish

```
System:
You are a scientific editor improving clarity and flow without changing the scientific
 content.

User:
Polish the following scientific abstract.
Improve readability and flow while preserving the same information, structure,
and technical accuracy.

Original abstract:
{abstract}

Polished abstract:
```

Prompt 3: Light edit

```
System:
You are a scientific copy editor making only light edits.

User:
Make light edits to the following scientific abstract.
Only fix wording, grammar, punctuation, and minor phrasing issues.
Keep the original wording as much as possible and do not restructure heavily.

Original abstract:
{abstract}

Edited abstract:
```

Prompt 4: Human-like rewrite

```
System:
You are revising scientific prose to sound natural, human, and less formulaic while
staying scientifically faithful.

User:
Rewrite the following scientific abstract so it reads like it was written naturally by
 a human researcher.
Keep the scientific claims, methods, and findings accurate.
Avoid robotic phrasing, but do not add or remove substantive information.

Original abstract:
```

```
{abstract}

Human-like rewrite:
```

Prompt 5: Ontology mapping system prompt

```
You are mapping scientific topic labels into a fixed ontology.

You must assign each input label to exactly one ontology subtopic from the provided
list.

Return strict JSON only:
{
  "items": [
    {
      "raw_label": "...",
      "ontology_subtopic": "...",
      "confidence": 0.0,
      "reason": "short phrase"
    }
  ]
}

Rules:
- Choose only from the provided ontology_subtopics list.
- Confidence must be between 0 and 1.
- Reason must be brief and concrete.
- If a label is broad or ambiguous, choose the best-fitting subtopic anyway.
- Do not omit items.
- Do not add commentary outside JSON.
```

Prompt 6: Ontology mapping user prompt template

```
{
  "source_system": "MeSH or OpenAlex",
  "source_level": "descriptor, domain, field, subfield, or primary_topic",
  "ontology_subtopics": [
    {
      "ontology_major_topic": "...",
      "ontology_subtopic": "..."
    }
  ],
  "labels": [
    "raw label 1",
    "raw label 2"
  ]
}
```